Hotel Booking Cancellation Prediction System

Dataset - Hotel_booking

🎯 Problem Statement

In the hospitality industry, booking cancellations are a major issue that leads to **revenue loss, poor resource planning, and inefficient operations**.

Hotels often struggle with:

- Overbooking or underutilization of rooms
- Last-minute cancellations
- Difficulty in forecasting demand

The challenge is:

> **Can we predict whether a hotel booking will be canceled or not based on booking details and customer information?**

🔍 Objective:

Build a **classification model** that predicts whether a booking will be **canceled (1) or not canceled (0)**.

💡 Business Impact:

- Helps hotels optimize **room allocation**
- Reduces revenue loss from cancellations
- Improves **pricing and overbooking strategies**
- Enhances customer targeting and retention

🎯 Target Variable

- **Column Name:** `is_canceled`
- **Description:** Indicates whether the booking was canceled
- **Type:** Binary
    - `1` → Canceled
    - `0` → Not Canceled

👉 This is a **binary classification problem**

📊 Dataset Overview

- Records: ~119,000 bookings
- Features: 30+ input features
- Type: Mixed (Numerical + Categorical)
- Domain: Hospitality / Revenue Management

📌 Column Information (Detailed Explanation)

1. **hotel**
    - Type: Categorical
    - Values: Resort Hotel, City Hotel

2. **is_canceled** *(Target)*
    - Binary indicator of cancellation

3. **lead_time**
    - Type: Numerical
    - Description: Days between booking and arrival
4. **arrival_date_year / month / week_number / day_of_month**
    - Type: Numerical / Categorical
5. **adults**
    - Number of adults
6. **children**
    - Number of children
7. **babies**
    - Number of babies
8. **country**

    - Country of origin
9. **stays_in_weekend_nights**
10. **stays_in_week_nights**
- Total nights stayed
11. **meal**
- Meal type booked
12. **market_segment**
- Segment (Online, Offline, Corporate, etc.)
13. **distribution_channel**
- Booking channel
14. **deposit_type**
- Values: No Deposit, Non Refund, Refundable
15. **adr (Average Daily Rate)**
- Type: Numerical
16. **previous_cancellations**
- Past cancellations by customer
17. **previous_bookings_not_canceled**
- Past successful bookings
18. **days_in_waiting_list**
- Time spent on waiting list
19. **reserved_room_type / assigned_room_type**
- Room types
20. **booking_changes**
- Number of booking modifications
21. **agent / company**
- Booking agent or company
22. **customer_type**
- Transient, Contract, Group, etc.